

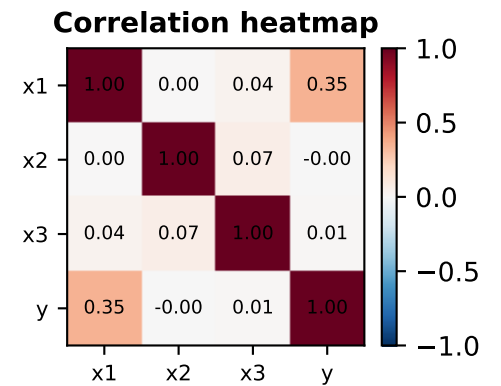
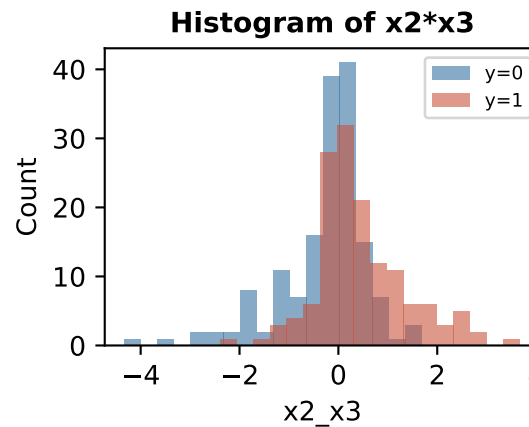
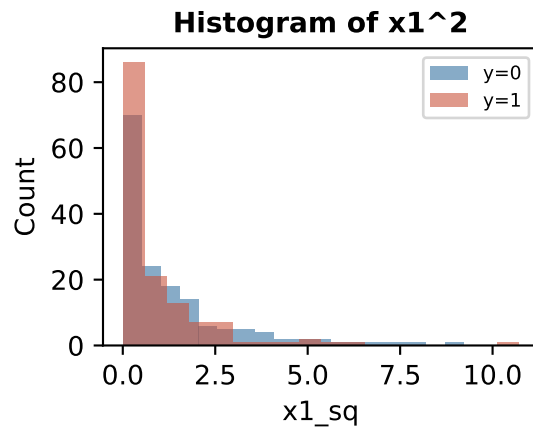
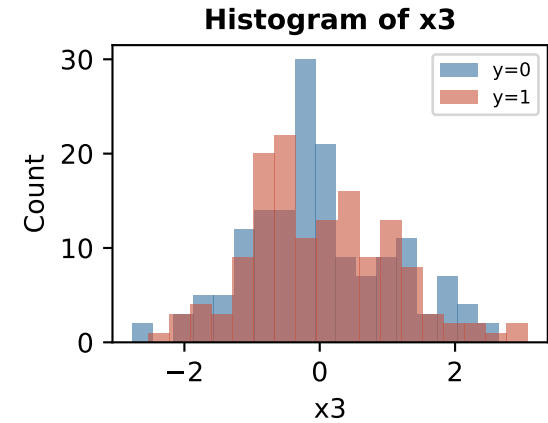
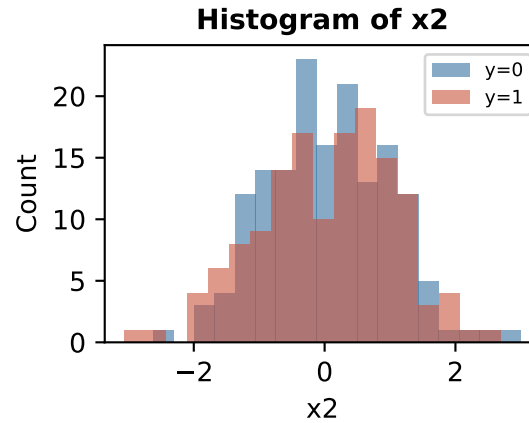
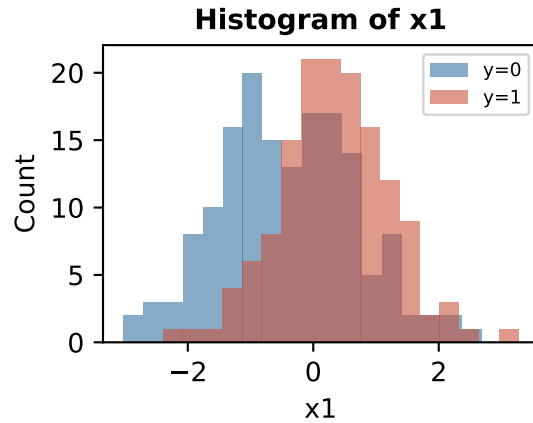
Data

Sample sizes

Training observations: 300
Test observations: 100
Class 0: 158 (52.7%)
Class 1: 142 (47.3%)
Predictors: x1, x2, x3

Summary statistics by class and total

Group	n	x1 mean	x1 sd	x2 mean	x2 sd	x3 mean	x3 sd
Total	300	-0.06	1.07	0.04	0.99	0.01	1.06
y=0	158	-0.41	1.09	0.04	0.95	0.00	1.07
y=1	142	0.33	0.90	0.04	1.04	0.02	1.05



Model, Method and Results

Model formula and libraries

Logistic model:

$$\text{logit}\{P(Y = 1 \mid x)\} = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 (x_2 x_3)$$

QDA model:

$$\delta_k(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \log(\pi_k)$$

Classify to $\text{argmax}_k \delta_k(x)$.

Libraries: statsmodels, scikit-learn, pandas, numpy, matplotlib

Assumptions

Logistic assumptions:

- independent binary outcomes
- log-odds are linear in selected engineered terms
- no perfect separation; sufficient sample size for SEs

QDA assumptions:

- predictors are approximately Gaussian within each class
- each class has its own covariance matrix
- observations are independent

Model Selection

Model	Specification	Log-loss	AUC	Accuracy	Precision	Recall	F1	FP/FN
Full logistic	all squares + interactions	0.520	0.821	0.767	0.739	0.791	0.764	12/9
Chosen logistic	x1, x1^2, x2*x3	0.502	0.836	0.767	0.750	0.767	0.759	11/10
Hierarchical logistic	x1,x2,x3,x1^2,x2*x3	0.508	0.823	0.722	0.696	0.744	0.719	14/11
QDA	class-specific covariance	0.512	0.848	0.767	0.750	0.767	0.759	11/10

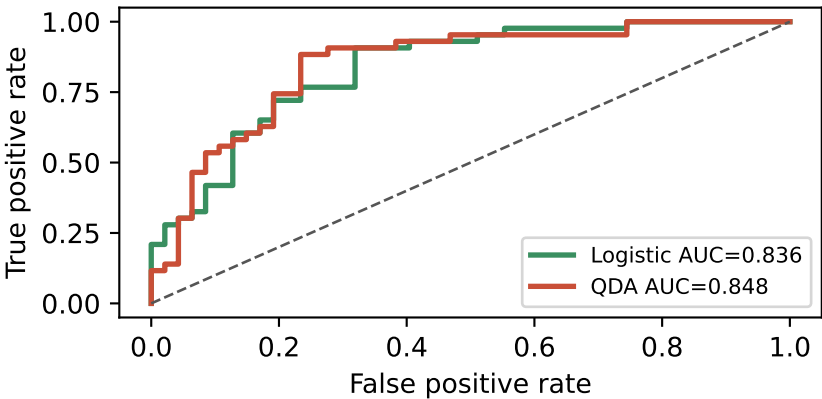
For model selection, the training data is split into fitting and validation parts; metrics are computed on the validation part.
The hierarchical logistic model is retained for comparison; final predictions use the logistic model with the lowest validation log-loss.

Logistic coefficients

Term	Coef	p
Intercept	0.053	0.753
x1	0.952	0.000
x1^2	-0.259	0.028
x2*x3	1.169	0.000

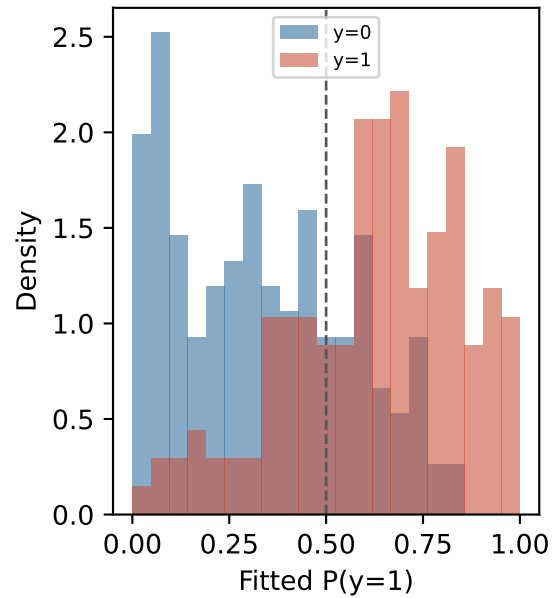
Insight: x1 raises class-1 odds; x1^2 bends that effect downward; x2*x3 is the key interaction signal.

ROC / AUC

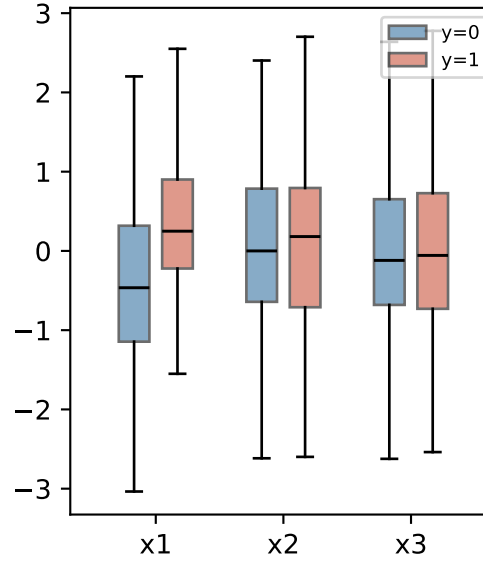


Visualization

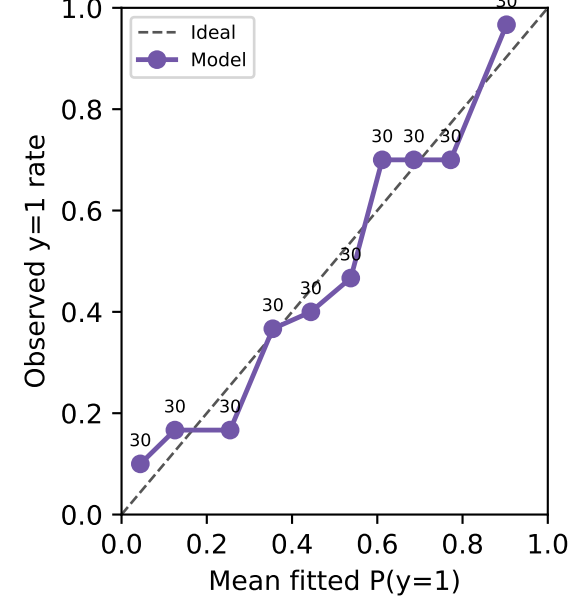
Predicted probability density



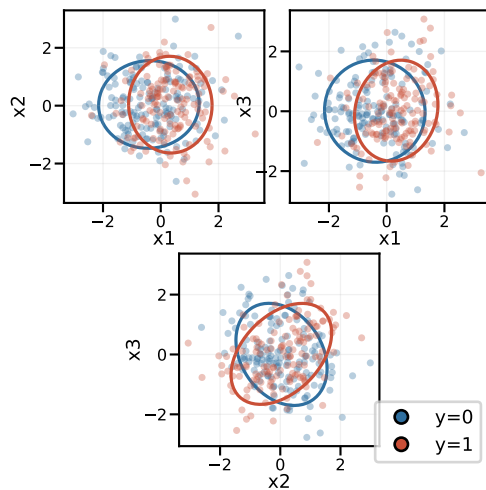
Covariates by class



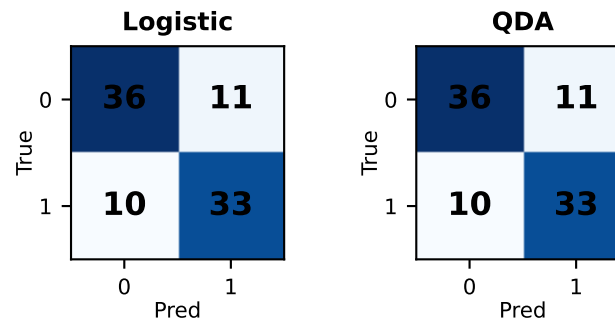
Reliability plot



Pairwise class clouds



Validation confusion matrices



$x_2 \times x_3$ probability surface

